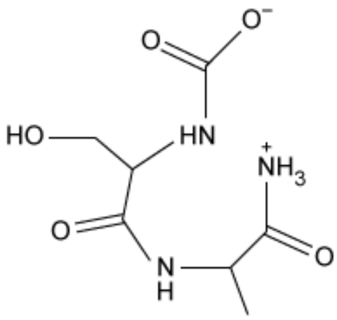
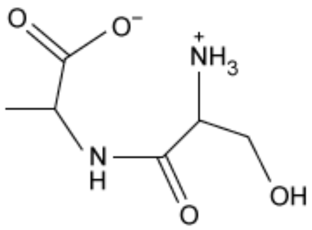
What is the structure of a dipeptide with the sequence Ala-Ser?

A.



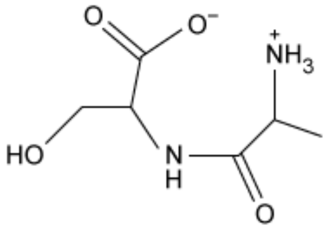
[4%]

B.



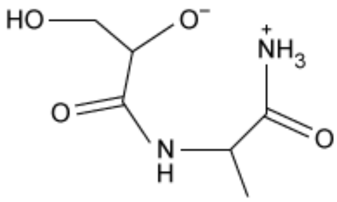
[36%]

C. ✔



[50%]

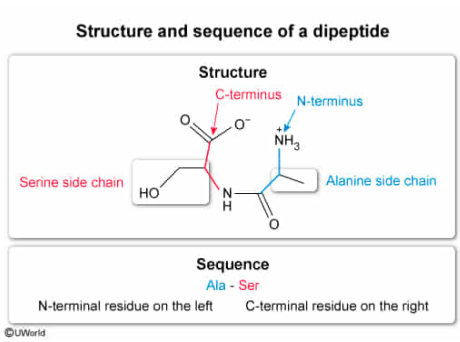
D.



[7%]

Correct
50% answered correctly

Explanation:



The backbone of an amino acid consists of an amino group (–NH_3^+, under physiological conditions) bonded to a carbon (the α-carbon), which is in turn bonded to a carboxyl group ($–COO^-$, under physiological conditions). The α-carbon is also bonded to a hydrogen atom (which is often omitted when drawn as a skeletal structure) and to a **side group** whose structure determines the identity of the amino acid.

The carboxyl group of one amino acid can react with the amino group of another to form a peptide bond that links the two amino acid **residues** of the dipeptide. The residue that retains a free amino group is called the amino-terminal or N-terminal residue, and the residue that retains a free carboxyl group is called the carboxy-terminal or C-terminal residue. By convention, the **sequence** of a peptide is always **written** with the **N-terminal residue on the left** and the **C-terminal residue on the right**, **regardless of how the structure of the peptide is drawn**.

The question asks for a peptide with the sequence Ala-Ser. Therefore, the correct structure must have an alanine residue at the N-terminus and a serine residue at the C-terminus. Only the structure shown in Choice C depicts a correct dipeptide backbone structure with an N-terminal alanine and a C-terminal serine.

(Choice A) This structure depicts the peptide Ser-Ala with an extra carboxyl group attached to the N-terminus and an extra amino group attached to the C-terminus.

(Choice B) This structure depicts the peptide Ser-Ala rather than Ala-Ser. Although the alanine residue is shown on the left side of the molecule, it is part of the C-terminal residue rather than the N-terminal residue.

(Choice D) In this structure, the carbonyl and α-carbons have their positions switched relative to where they should be. In each residue, the α-carbon (to which the side chain is linked) should be *between* the nitrogen and the carbonyl carbon.

Educational objective:
The backbone of an amino acid consists of an amino group linked to an α-carbon, which is in turn linked to a carboxyl group. The α-carbon is also linked to a side chain that determines the identity of the amino acid.

Time spent:0 sec
Subject:
Biochemistry

QID:403677
System:
Amino Acids and Proteins